

The Science-based Playbook of **SaaS** **Optimization**

Ariyh“

Boost your performance using the latest scientific discoveries in SaaS marketing. No opinions, no sketchy data, no fluff.



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About this playbook

This SaaS Optimization playbook is a practical guide on how to apply the latest scientific evidence to increase your SaaS revenue and profits.

All science covered in this playbook has survived rigorous statistical analyses, been peer-reviewed (i.e. picked apart and challenged by other scientists), and been published in a high-quality scientific journals.

The Science-based Playbook of SaaS Optimization is not:

- Built on opinions or “advice from experts”. This is science.
- Generic, vague information. It’s actionable and straight to the point.
- A beginner's guide to SaaS marketing. These are advanced optimization techniques.

Remember, the context of your SaaS will change how effective these techniques are for you (for better, or for worse). So whenever possible A/B test these recommendations before fully implementing them. A culture of A/B testing brings many other benefits. On average, startups that A/B test constantly perform 30% to 100% better than those that don't.

If you find this playbook useful, you'll probably find [Ariyh's Insights Platform](#) useful too. Members get full access to 200+ science-based insights, real-world case studies, and exclusive playbooks.

If you're interested in optimizing your promotions and understanding your price down to the digit (e.g. \$20 vs \$19.99), you can refer to the [The Science-based Playbook of Pricing & Promotions](#).

Alright, it's time to dive in. Hope you'll enjoy this as much as I enjoyed putting it together!

Thomas McKinlay,
Founder of Ariyh

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What Ariyh readers say

“ In just 10min of skimming through the playbook, I found 8 tactics that we're already on the way of optimizing.



Baptiste Nicolas - Founder
VidJet

“ An incredible source of best practices and ideas for marketing experimentation. When I need to test or optimize something, I start with Ariyh.



Helen Ivashko - Sr. Growth Marketing Manager
Amazon

“ An insanely valuable resource for any SaaS marketer worth their salt.



Diane Wiredu - Founder
Lion Words

“ Ariyh does an excellent job at translating our academic research into clear and actionable advice.



Norbert Schwarz, Provost Professor
Marshall School of Business

“ Not only does it cover the latest, most relevant studies - it does so evenly, with both strengths and flaws of the research.



Richard Shotton, Bestselling author
The Choice Factory

“ I used to just rely on advice from our agency. Mistake, everything was bland and ineffective. This playbook has been a blessing.



Can Erdoğan - Founder
Real Metric Analytics

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Some of the institutions we should be grateful to for the research covered in this playbook:





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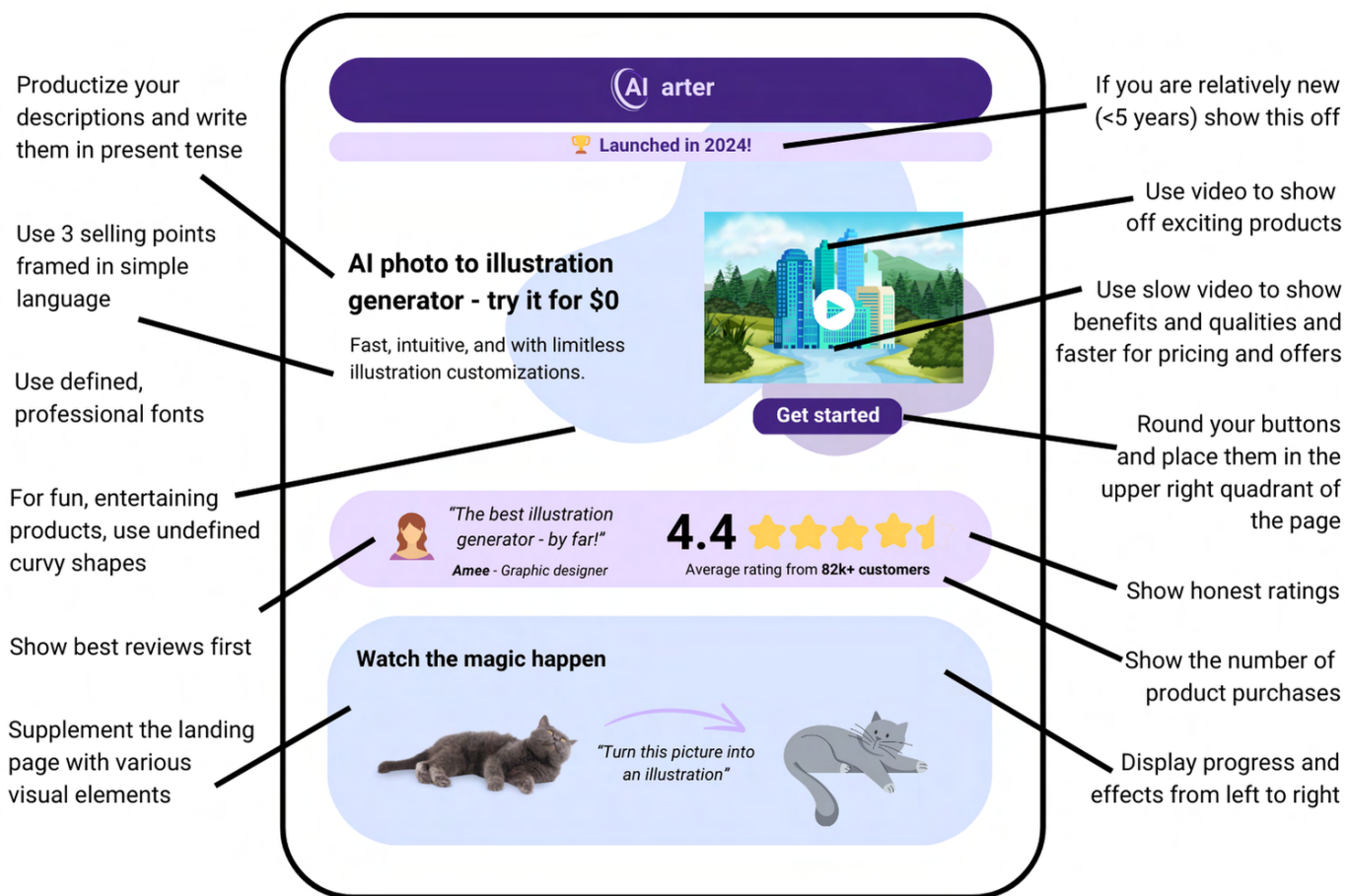
“ 1.

Brand and messaging optimization

The optimal SaaS homepage

This is an example mockup of a SaaS homepage optimized with some of the latest marketing research.

The exact components and messages will depend on your branding, product and goals (e.g. a product that needs to be perceived as fun and exciting is different from one that needs to be perceived as reliable and dependable). We dive into each of these in this section.



Productize what your software does and be specific



Recommendation

- Frame your software as an end-to-end 'product' with a clear and specific outcome, message, and price, rather than a generic service (e.g. don't sell an "image editor". Sell "AI video wizz: an automated background remover, for \$79/year").
- People will find it easier to understand the value they get, so they'll be more likely to buy, pay a higher price, and be happier with the outcome.



How it works

- When people find something difficult to visualize, it feels like a riskier purchase, so they are less likely to buy.
- A clearly defined product is easier to picture and is more likely to meet the expectations of the buyer.



Research

Wirtz, J., Fritze, M. P., Jaakkola, E., Gelbrich, K., & Hartley, N. [Service products and productization](#). *Journal of Business Research* (September 2021).

Universities: National University of Singapore; University of Cologne; Turku School of Economics, University of Turku; Catholic University Eichstaett-Ingolstadt; and University of Queensland.

[Full Ariyh research summary](#)

PRO TIP: Use the present tense (vs past tense)

The present tense implies something current and more universal, seeming more certain (e.g. "It's very popular with engineers worldwide" and not "Since 2019 it has been very popular with engineers worldwide").

PRO TIP: Show off that you are new

People prefer high-tech or innovative products when they are from new brands (vs established, older competitors). If you are relatively new (<5 years) show this off (e.g. "Established in 2024"), especially if your SaaS is highly innovative (e.g. using AI). If you're not, avoid mentioning it front and center.

Highlight only your top 3 key benefits



Recommendation

- Showcase your 3 best benefits or features (e.g. Fully customizable, World-class cybersecurity, Guaranteed uptime), not more, not less. You will be more persuasive.
- The effect follows a n pattern, with 1 or 2, as well as 4 or 5 (and beyond), being less persuasive than 3. In the study, 3 claims were 10.4% more persuasive than 4.
- Hold off on describing your other features in most of your messaging. Instead, keep them for a specific “features” page or section.



How it works

- Using 3 positive claims shows a positive pattern. Once people feel they have identified a pattern, they are more likely to make a decision.
- Any claims beyond that start to make people skeptical, so they backfire.



Research

Shu, S. B., & Carlson, K. A. [When three charms but four alarms: Identifying the optimal number of claims in persuasion settings](#). *Journal of Marketing* (January 2014).

Universities: UCLA Anderson School of Management, University of California & McDonough School of Business, Georgetown University.

[Full Ariyh research summary](#)

PRO TIP: Don't showcase all of your good features

When you showcase your best features alongside your average or good ones, you're [diluting your best ones and you'll be perceived as worse overall](#). On the homepage only highlight the parts of your product that set you apart from the competition and are really worth mentioning.

PRO TIP: When describing your products and features, use simple language

Unfamiliar or [technical language is difficult to understand](#) and requires mental effort (e.g. “easy to use” compared to “frictionless toolstack and workflow integration”). If people don't understand a product description they're also less inclined to trust it and buy it.



Page design and visuals

Design to appear effective and reliable or fun and exciting



Recommendation

- If people buy your product because it's effective, safe, and reliable (e.g. fintech), use designs that look and feel structured (e.g. geometrical shapes, straight lines, symmetrical layouts).
- If people buy your product because it is exciting, fun, and enjoyable (e.g. easy-to-use creative tools), use designs that look and feel unstructured (e.g. incomplete shapes, rounded lines, unpredictable layouts).
- These design principles apply to all your designs (e.g. logo, ads), not only the homepage.



How it works

- Structured designs (e.g. straight lines) feel organized, stable, and predictable. So the brand seems effective and reliable.
- Unstructured designs (e.g. incomplete shapes) feel surprising, dynamic, and unpredictable. So the brand seems fun and exciting.
- People constantly look for signals that back up a company's claims. Aligned design elements are a key way to do that.



Research

Affonso, F. M., & Janiszewski, C. [Marketing by Design: The Influence of Perceptual Structure on Brand Performance](#). *Journal of Marketing* (November 2022).

Universities: Oklahoma State University & University of Florida

[Full Ariyh research summary](#)

PRO TIP: Use colors that fit your brand

Use colors that [align with your positioning](#). Red is associated with excitement (e.g. use for an event ticketing platform), blue with competence (e.g. use for accounting software), and black with sophistication (e.g. use for an engineering project management tool).

PRO TIP: B2B audiences prefer messages in cold and dark colors

B2B audiences tend to prefer [messages in cold and dark colors](#) (e.g. blue, dark green, purple), as they signal your competence. For B2C audiences, use warm colors (e.g. yellow, bright green, red) to simulate emotions.

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Adapt your layout depending on your type of SaaS



Recommendation

- If you're an already well-known and trusted brand (e.g. Adobe Photoshop) give more descriptive details, focus on technical specifications, use a comparison matrix, and offer clear recommendations of how to use it (e.g. use cases and scenarios).
- If your brand is less trusted or not yet well known (e.g. new AI animation generator): focus on entertaining elements (e.g. dynamic visuals, illustrative animations, and GIFs), use interactive elements (e.g. live demos and free trials), and strong social proof (e.g. testimonials or specific case studies).
- If your SaaS is experience-related (e.g. travel booking system): use conversational language (e.g. "you," "your"), include a lifestyle picture or product video, and make all relevant information instantly available (e.g. without the need for users to click "show more" buttons).



How it works

- People's decision-making differs for different types of products. Aligning your layouts and designs to that increases sales.
- The right type of design elements can make your SaaS seem more personalized, user-friendly, and trustworthy.



Research

Bleier, A., Harmeling, C. M., & Palmatier, R. W. [Creating Effective Online Customer Experiences](#). *Journal of Marketing* (December 2018).

Universities: Frankfurt School of Finance & Management, Florida State University, and University of Washington

PRO TIP: Design your page to be moderately complex

Include a mix of core elements (e.g. description of the main functionalities) and non-core elements (e.g., promotions, testimonials, case studies) to [draw more attention and focus to the product](#). You want the page to look moderately complex, but not too complex (which would overwhelm the viewer).

PRO TIP: Display pictures of your support team

If customer support is an important element for your SaaS (e.g. hotel booking platform), display [pictures and names of members of your support team](#). This will create the sense that any inquiries will be taken care of promptly and people will be more likely to convert.

Make your CTA buttons rounded



Recommendation

- Use curved edges or an oval shape (instead of straight angles) on your positive CTA buttons (e.g. “sign up” or “buy”).
- Across 8 experiments (websites, ads, eye-tracking lab tests) they received 17% to 55% higher click rates and looked at the button 29% longer.
- For buttons centered on avoidance-related actions (e.g. “Opt-out”) or negative framing (e.g. declining an offer), you can use either rounded or straight edges (like 90-degree angles) for your buttons.



How it works

- People prefer round shapes because they associate roundness with friendliness and harmony. They are more likely to approach (click) such friendly, attractive objects.
- People associate sharp angles with threats and toughness and are less likely to approach (click) them.



Research

Biswas, D., Abell, A., & Chacko, R. [Curvy Digital Marketing Designs: Virtual Elements with Rounded Shapes Enhance Online Click-Through Rates](#). *Journal of Consumer Research* (December 2023).

Universities: University of South Florida, University of Tennessee & Neighborly

[Full Ariyh research summary](#)

PRO TIP: Use a machine-written, not a handwritten font

Use a [machine-written-looking font](#) for your SaaS (e.g. Arial, Roboto, Times New Roman) to appear precise and reliable (unless you're a fun, creative SaaS, in which case you can try using a handwritten-looking one while making sure it's easily readable).

Include a CTA button in the upper right quadrant



Recommendation

- Position a CTA button (e.g. “Sign up”, “Try it”) in the lower part of the upper right quadrant of the homepage. This shouldn’t be the only one. You should likely also have CTAs in other places too (e.g. lower on your page).
- Across multiple websites, eye-tracking data found that people clicked on CTA buttons 1-1.5 seconds quicker when they were in the upper right quadrant.
- Place over 70% of the essential content (e.g. an explanation about the product, a demo video, a CTA button) above the fold of a diagonal line from the top-left to the bottom-right of the page.
- Create a clean and minimal design that minimizes distractions and guides the user towards the CTA naturally.



How it works

- People’s natural eye movement follows the top-left to the bottom-right diagonal pattern (the Gutenberg pattern).
- Shaping your homepage to follow this pattern creates a straightforward visual hierarchy, reduces cognitive load, and improves the user experience.



Research

Hernandez, A., & Resnick, M. L. [Placement of Call to Action Buttons for Higher Website Conversion and Acquisition: An Eye Tracking Study](#). *Proceedings Of The Human Factors And Ergonomics Society* (September 2013).

Universities: Bentley University

Show the “before” on the left and the “after” on the right



Recommendation

- Place images that represent the past on the left and images that represent the future on the right (e.g. the result of your solution in action - a disorganized email inbox on the left and organized inbox on the right).
- People will like your product more.
- For right-to-left reading cultures (e.g. Hebrew), reverse the positioning with past-related images on the right and future-related images on the left.



How it works

- People naturally associate the past with the left and the future with the right in left-to-right reading cultures, making this layout much more intuitive. This is reversed for right-to-left reading cultures.
- Placing an image of the future on the right feels natural and promises a future state (e.g. organized messages). Placing it on the left would intuitively represent the “before.”



Research

Chae, B., & Hoegg, J. The [Future Looks “Right”](#): Effects of the Horizontal Location of Advertising Images on [Product Attribute](#). *Journal of Consumer Research* (January 2013).

Universities: University of British Columbia

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Use video to show off exciting and enjoyable software



Recommendation

- If your software is fun and enjoyable (e.g. mash-up image generator tool), use plenty of video or GIFs, rather than static pictures, to increase how likely people are to choose it by up to 79%.
- If your SaaS is particularly functional (e.g. web hosting platform) investing in video is not necessary, unless it's needed to explain complex features.
- Focus on showing how to use the product and what kind of benefits people will receive by making the purchase.



How it works

- Dynamic formats (e.g. video, slideshow, GIFs) present information in a more lifelike way than static images.
- This makes it easier to imagine using and enjoying the product and people become more likely to buy it.



Research

Roggeveen, A. L., Grewal, D., Townsend, C., & Krishnan, R. [The impact of dynamic presentation format on consumer preferences for hedonic products and services](#). *Journal of Marketing* (November 2015).

Universities: Babson College and University of Miami

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How to use video to show quality versus features



Recommendation

- Use slower video when highlighting product benefits and quality (e.g. “Unlock new levels of creativity with The Creation Pack”).
- Use faster video when highlighting product features and price (e.g. “The Creation Pack offers a range of video editing tools, now for \$65 per month”).



How it works

- When something moves slowly people perceive it as further away, when it moves fast it feels closer.
- People tend to judge objects that are further away in more abstract terms and at a higher level (the benefits and the “why” of the product).
- People judge objects that are closer in more practical, lower-level terms (whether it’s actually achievable and “how”).
- In addition, people are fast at processing characteristics of products (e.g. price, features, realizing that something is a good deal (“50% off”)), but slower at processing the meaning of them (the benefits, the “why”, “I would be happy to use that because...”).



Research

Yoon, S., Bang, H., Choi, D., & Kim, K. [Slow versus fast: How speed-induced construal affects perceptions of advertising messages](#). *International Journal of Advertising* (May 2020).

Universities: Babson College and University of Miami

[Full Ariyh research summary](#)



Social proof and reviews

Show your views or sales numbers



Recommendation

- Showcase your number of sales (e.g. “Trusted by 35,000+ analysts”) to boost people’s trust and purchase intentions by up to 53.5%.
- Purchases are a stronger signal than views, but views are much stronger than nothing. For example, “156 views” increased purchase intentions by 30%.
- So when you are starting out, display your number of views, then once you cross a few hundred sales switch to displaying your number of purchases.



How it works

- Product views or purchases by others are a signal that people use to make assumptions about the product’s quality. If others have checked it out or bought it, it must be good.



Research

Das, G., Spence, M. T., & Agarwal, J. [Social Selling Cues: The Dynamics of Posting Numbers Viewed and Bought on Customers’ Purchase Intentions](#). *International Journal of Research in Marketing* (January 2021).

Universities: Indian Institute of Management Bangalore; Bond University; and Haskayne School of Business, University of Calgary.

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PRO TIP: State how many users you expect to have in the future

If you are starting a marketplace or two-sided platform (e.g. a forum that requires enough members to be useful) and don’t yet have many users or customers, state [how many users you expect to join](#) (e.g. “We expect 30,000 users to sign up by June”). People will be more likely to sign up.

Aim for and display a great but imperfect average rating



Recommendation

- Aim to show an average review rating of around 4.5 stars, not a perfect 5-star score. Cultivate this rating on external platforms (e.g. G2, Capterra) and in what you show on your own website.
- Average review ratings increase sales as they go up and peak at between 4 and 4.5 stars, then slightly lower them again.
- So while you should aim for an excellent rating (5 stars is better than 3), it's alright to have some negative reviews that lower your average.



How it works

- If the average rating for a product is extremely high, people tend to think that they are too good to be true. They suspect that the company could have manipulated them.
- Instead, when some negative reviews are present but the rating is still good (4 to 4.5 stars), people assume that the reviews honestly discussed both positive and negative aspects, but the product still came out as great.



Research

Maslowska, E., Malthouse, E. C., & Bernritter, S. F. [Too good to be true: the role of online reviews' features in probability to buy](#). *International Journal of Advertising* (June 2016).

Universities: University of Illinois Urbana-Champaign, Northwestern University, King's Business School, King's College London.

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PRO TIP: [Some negative reviews actually help you](#)

A few [negative reviews will boost your sales](#) if they're irrelevant to most customers (e.g. a certain language is not supported, they don't like the color of your logo). Some negative reviews enforce the legitimacy of the positive ones.

Curate the first review or testimonial you show



Recommendation

- When choosing which user testimonials to display first, always opt for ones highlighting the core features and unique selling points of your product.
- If you have a review section, or on external review websites (e.g. Capterra, G2) do everything you ethically can to receive the first reviews that are positive. You can do so by sharing early access to the product with your loyal existing users (e.g. your beta testers) and offering them extended free access if they leave you an honest review.
- Most likely, they will leave you a positive one (if they don't, it's a serious signal you need to improve your product).



How it works

- A negative first review leads to a chain reaction of fewer sales, fewer reviews, and more negative reviews.
- The opposite holds true for seeing a positive review.



Research

Park, S., Shin, W., & Xie, J. [The fateful first consumer review](#). *Marketing Science* (February 2021).

Universities: Darla Moore School of Business, University of South Carolina and Warrington College of Business, University of Florida

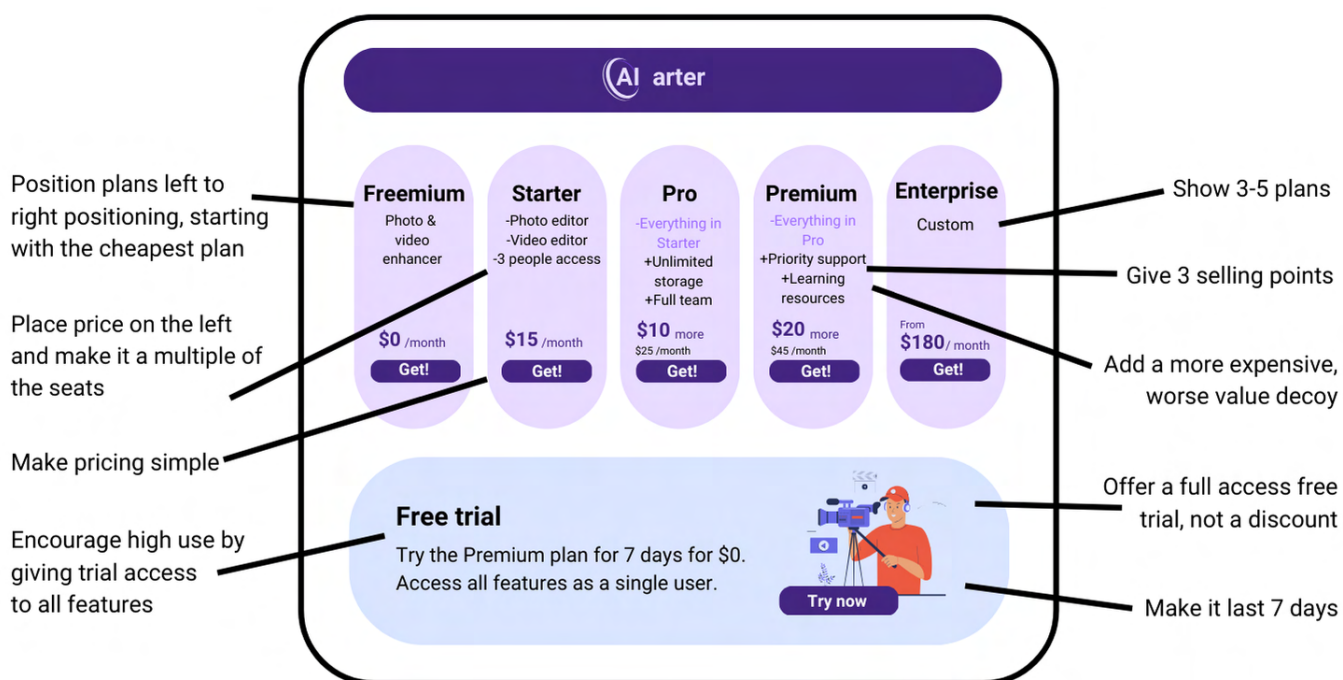
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“ 2.

Pricing plan optimization

Below is an example mockup of a SaaS Plans page optimized with the most impactful insights from the latest marketing research.

The number of plans, as well as pricing strategy (e.g. flat vs. per seat), will differ across different products types. Choose what makes the most sense in your business and combine it with the learnings you have of your own business. A/B test whenever possible.





The Plans

Offer between 3 and 5 plans



Recommendation

- To ease user decision-making, aim to offer between 3 and 5 plans.
- Offer a basic plan as a low-cost entry option, a standard plan as a balance between cost and functionality, and a premium plan focused on capturing a high-end market segment.
- Make sure the middle option is well-defined as a balance between the basic and high-end options - people will be most drawn toward this one.
- You can consider an optional enterprise plan for large organizations with custom needs.
- Finally, if you opt for a freemium model (more on this below), you can include a free plan.



How it works

- People tend to avoid options that are perceived as extreme or risky (e.g. the most expensive one) and prefer the middle or moderate options, thus most opt for the Standard plan.
- This effect is driven by people wanting to make a decision that is safer, reducing feelings of regret.



Research

Simonson, I., & Tversky, A. [Choice in Context: Tradeoff Contrast and Extremeness Aversion](#). *Journal of Marketing Research* (1992).

Universities: Haas School of Business, University of California Berkeley, and Stanford University.

Consider turning a middle plan into a decoy



Recommendation

- When presenting multiple plans, you can turn one of the plans into a decoy option that is slightly better but significantly pricier (e.g. make your middle plan a decoy).
- The goal is not to sell the decoy plan but to make the other plans (lower priced and higher priced) more attractive.
- Be sure to use just-below prices instead of rounded pricing on your main plan too (e.g. \$49.99 vs. \$50). A decoy option led people to buy the target (e.g. standard plan) option 62% of the time when using a just-below price (vs. 49% when using rounded prices).



How it works

- The decoy effect relies on presenting an option that makes another choice seem superior by comparison.
- By introducing a decoy, precise pricing helps people identify the superior value of the target option, which then leads them to choose it over others.



Research

Cui, Y. G., Kim, S. S., & Kim, J. [Impact of preciseness of price presentation on the magnitude of compromise and decoy effects.](#) *Journal of Business Research* (August 2021).

Universities: Auckland University of Technology and The Hong Kong Polytechnic University.

Order plans left to right and place your price on the left side



Recommendation

- Display your plans left to right - from the cheapest to the most expensive one.
- Display product prices on the left side of each specific plan image or description (vs to the right) to make them appear more affordable.
- People assigned up to 9.49% higher prices for products (e.g. clocks, wine, lamps, staplers) that were placed on their right-hand side.



How it works

- When people process items from left to right, they start with the smaller values on the left (e.g., 1, 2, 3, etc.), which influences people's perceptions of the prices displayed on the left as being "lower".
- So people associate smaller numerical values with the left side of a display, such layout flows more naturally indicating progression.



Research

Cai, F., Shen, H., & Hui, M. K. [The Effect of Location on Price Estimation: Understanding Number-Location and Number-Order Associations](#). *Journal of Marketing Research* (October 2012).

Universities: Shanghai Jiao Tong University & Chinese University of Hong Kong

PRO TIP: Display your price below the plans

Position the price below the features list and above the CTA button. This makes people perceive the price as lower and makes them more likely to subscribe. The "down = less" metaphor is deeply ingrained in people, so a lower vertical position makes the price seem smaller.

Focus on the price difference of premium plans (e.g. \$15 more)



Recommendation

- When displaying prices of different plan option, upgrades, or add-ons (e.g. Starter to Premium plan), focus on the price difference (e.g. \$10 more for Premium) not the total price (e.g. \$15 Starter vs \$25 for Premium).
- Your premium plan will seem cheaper, because \$10 (the difference) is cheaper than \$25 (the total price), so people will be more likely to buy it.
- This does not work for more expensive, high-involvement purchases, as people will stop and properly calculate the prices themselves.



How it works

- People think of prices as they're shown to them. For example, "\$5 per month, when buying the annual plan" makes people focus on the monthly, rather than the total yearly amount, making people consider the total cost as cheaper.
- Similarly, when people see the price difference for a premium plan, they focus on that number more than the total cost.



Research

Allard, T., Hardisty, D. J., & Griffin, D. [When "More" Seems Like Less: Differential Price Framing Increases the Choice Share of Higher-Priced Options](#). *Journal of Marketing Research* (July 2019).

Universities: Nanyang Technological University, University of British Columbia

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Make your price simpler than your competition



Recommendation

- Avoid complex pricing, which is difficult to understand at a glance (when price components have many different odd numbers (e.g. \$2.7/per seat, \$0.55/per GB of storage) and require different calculation types (e.g. \$20/month service fee plus 10 x \$3 seats with 20% entry discount).
- People tend to choose simple price options. In experiments, even when a plan was more expensive, (€35 vs €30 per month), 65.9% still chose the simple option.
- Stick to even endings, consider flat-rate prices (more on this below), and ensure customers only need to make simple calculations to get the final price (e.g. show prices with discounts applied).



How it works

- When the price is complex, people may not put in the effort, and simply assume the price is higher than a simpler option (even if it isn't).
- People may also feel like the company is being unfair and making it hard to figure out the cost.
- When the price is simple, people feel that the company is being more fair and transparent than competitors, so they are more likely to choose it.



Research

Homburg, C., Totzek, D., & Krämer, M. [How price complexity takes its toll: The neglected role of a simplicity bias and fairness in price evaluations](#). *Journal of Business Research* (June 2013).

Universities: University of Melbourne, University of Mannheim, and University of Passau

[Full Ariyh research summary](#)

Make your prices easy to divide or multiply



Recommendation

- List the number of product attributes or the number of users the product plan is intended for as a multiple of the price (e.g. \$36 team plan for 3 or 6 users), so it's easy to calculate.
- For instance, you can set a price for:
 - Team subscriptions at \$15 for 3 or 5 users (\$5 or \$3 each)
 - Cloud storage at \$5 for 50GB or \$10 for 100GB.
 - Usage limits for, for example, the \$5 plan allow generating 25 AI images.
- You can also use this in your promotions, for example, "save \$12" and include a tagline of 4 reasons why to choose your product.



How it works

- People prefer to see numbers that are easily cognitively processed, especially resulting from common arithmetic problems.
- Exposure to multiples of a number (e.g., 2 and 6) immediately activates the sum (e.g., 8) along with their product (e.g., 12) which makes it easier for us to process and leads to greater liking.



Research

King, D., & Janiszewski, C. [The Sources and Consequences of the Fluent Processing of Numbers.](#) *Journal of Marketing Research* (April 2011).

Universities: National University of Singapore & University of Florida

Use a combined flat + usage pricing if you have advanced analytics



Recommendation

- If you have sophisticated enough analytics and can constantly monitor usage, a fixed fee (e.g. \$19/month) and pay-per-use (e.g. \$10 per 1,000 visits) combination is more profitable.
- In this situation, two-part pricing plans lead to 36% and 49% higher profits compared to just pay-per-use and flat-rate plans, respectively.
- Use data to determine the optimal balance between the fixed fee and the marginal usage fee to maximize profits.



How it works

- Customers adjust their usage based on marginal price changes.
- Two-part pricing plans efficiently capture surplus by separating the decision to subscribe from the decision on usage. The fixed fee secures a baseline revenue while the marginal price adjusts to their usage levels.



Research

Schlereth, C., Skiera, B., & Wolk, A. [Measuring Consumers' Preferences for Metered Pricing of Services](#). *Journal of Service Research* (December 2011).

Universities: Goethe-University Frankfurt.

Use flat-rates as an effective straightforward approach



Recommendation

- If you prefer a simpler approach, want to reduce complexity (e.g. because other parts of your product are complex), and want to increase upselling or usage, stick with a flat fee (e.g. the same price no matter the number of seats and the usage intensity).
- In experiments in this study, 68% of people chose the flat-rate plan when the pay-per-use plan would have cost the same, and 23% chose the flat-rate plan even when it was 20% more expensive than pay-per-use.



How it works

- People prefer flat-rate plans over pay-per-use plans, even if that means they pay more. Both B2C and B2B customers have this bias.
- People tend to overestimate how much they will use a product. It also feels convenient, and simple and gives a sense of insurance that people will not be charged more than they expect.



Research

Kienzler, M., Kowalkowski, C., & Kindström, D. [Purchasing professionals and the flat-rate bias: Effects of price premiums, past usage, and relational ties on price plan choice](#). *Journal of Business Research* (April 2021).

Universities: Linköping University and Hanken School of Economics

[Full Ariyh research summary](#)

PRO TIP: Do not overwhelm with too many choices at once

If you must offer detailed customizations (e.g. number of team members, data allowance), try not to ask people to make this [choice at the same time](#) as choosing a plan. Ask them as a follow-up question (e.g. as a dropdown, a follow-up page), or clearly group the choice as separate.

Need more pricing insights?

If you need even more insights into setting your prices, you can refer to

[The Science-based Playbook of Pricing & Promotions](#). It's designed to help you:

- Learn how to reliably survey (potential) customers to understand how much they're willing to pay.
- Understand when to use rounded (\$20), just-below (\$19.99), or precise (\$22.37) prices.
- Set up promotions and discounts that are more likely to convert and boost your sales.

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Free Trials

Offer a high-quality free trial



Recommendation

- Offer a high-quality free trial or sample of your product.
- The free trial should give access to all your features and your content. Instead of limiting the number of features, limit the usage of each (e.g. only 3 downloads during the free trial vs unlimited downloads when paying).
- Designing your free trial or free samples this way increased revenue by 7.7% in experiments.



How it works

- When a product is high-quality, the sample 'taster' creates trust and is the final push to make the purchase.
- Be mindful that for low-quality products, free trials can reduce sales (as people see it's not worth their money).



Research

Li, H., Jain, S., & Kannan, P. K. [Optimal design of free samples for digital products and services.](#) *Journal of Marketing Research* (April 2019).

Universities: The Ohio State University, University of Texas at Dallas, and University of Maryland.

[Full Ariyh research summary](#)

7 days is the optimal free trial length



Recommendation

- Make your free trial last 7 days (instead of 14 or the standard 30). People will be more likely to convert and stay longer, giving you a higher lifetime value (LTV).
- In a large-scale experiment, 7-day long free trials (vs 30 days) led to 5.6% higher conversions, 6.4% better retention, and 7.9% higher revenue.
- You could test even shorter trials (e.g. 24 hours, 3 days) as long they give enough time to actually try the product, but these were not tested in this study.



How it works

- The more intensively people use a product during a free trial the more likely they are to buy it.
- Shorter trials create a sense of urgency and lead to more intense product use.
- A lack of urgency of long trials means that people feel less pressure to use the product and are more likely to forget about it.
- A shorter, but more intense, use of the product in a short trial leads to higher conversions.



Research

Yoganarasimhan, H., Barzegary, E., & Pani, A. [Design and evaluation of optimal free trials](#). *Management Science* (August 2022).

Universities: University of Washington

[Full Ariyh research summary](#)

Encourage high usage during the trial period



Recommendation

- Focus on encouraging high product usage during the 7-day trial period.
- Provide a range of features (e.g. grant access to all functionalities the user might need), a comprehensive onboarding flow, dedicated customer support teams, or tailored recommendations to further enhance user experience and involvement.
- Trials with high usage (30% to 50% above average) lead to more adopters and also reduce any negative word of mouth.
- This should boost the likelihood of conversion into a paid subscription too.



How it works

- Higher usage helps to understand the value of the product and boosts familiarity and trust.
- High usage during trial periods also creates a sense of commitment, which leads to a higher likelihood of converting to paid.



Research

Foubert, B., & Gijsbrechts, E. [Try It, You'll Like It—Or Will You? The Perils of Early Free- Trial Promotions for High-Tech Service Adoption](#). *Marketing Science* (May 2016).

Universities: Maastricht University and Tilburg University.

PRO TIP: Create a sense of “closure”

A [sense of “closure”](#) can motivate people to upgrade from a free to a premium membership. Lean into gamification elements such as awarding badges for users who upgrade to paid or showing users their progress in exploring the app while highlighting how upgrading will help them unlock new milestones.

Give free-trial extensions instead of discounts

Recommendation

- If a user fails to convert at the end of the free trial, offer a free trial extension rather than a low, discounted price (e.g. "Need more time to decide? Enjoy 7 more days of free access on us")
- People will perceive the service as more valuable, leading to a higher willingness to pay and a higher conversion likelihood.



How it works

- A discount makes the price of the product lower, thus seemingly diminishing its value (e.g. people were willing to pay up to 50.1% less for a product when it was offered at a discount rate vs. free).
- Free trials do not "diminish the value" of the product.
- And people are more likely to appreciate it.



Research

Palmeira, M. M., & Srivastava, J. [Free Offer ≠ Cheap Product: A Selective Accessibility Account on the Valuation of Free Offers](#). *Journal of Consumer Research* (December 2013).

Universities: Monash University and Temple University.



Freemium

Limit usage, not features



Recommendation

- If you use a freemium business model, allow free users to try any of your features (or as many as possible). Instead, limit the usage of each feature at a low level (e.g. only 1 export per month). This is more effective than limiting access to features.
- In experiments, this approach boosted subscriptions by up to 31%.



How it works

- Being able to sample all types of features allows users to better understand how willing they are to pay for your product.
- Limiting the access then encourages them to start a paid subscription.



Research

Aral, S., & Dhillon, P. S. [Digital Paywall Design: Implications for Content Demand & Subscriptions.](#) *Management Science* (August 2020).

Universities: MIT Sloan School of Management

[Full Ariyh research summary](#)

PRO TIP: Frame your freemium as costing \$0

Frame your freemium as costing \$0, rather than 'free'. Using "\$0" or "at no cost" reminds people they are not losing money, rather than gaining something "free". That increases conversions.

Add a decoy plan to make your freemium users upgrade



Recommendation

- In a freemium model, offering only a free plan and one paid plan is suboptimal. You should have a third, decoy, paid plan.
- The decoy plan should be either:
 - Much more expensive than the main plan, with more features and higher usage limits.
 - Slightly cheaper, 70%-80% the price of the main plan, with only slightly fewer features or usage limits.
- In experiments this drew more attention to the main plan, making it seem like a good deal and boosting sales of it between 16% and 23%.



How it works

- People are likely to choose the 'middle option', rather than an extreme one (most cheap or most expensive). It seems like a good compromise.
- People can also experience the attraction effect. Seeing a clearly 'inferior option' (close in price but quite far from being as good as the main option) makes people more likely to choose the 'dominating option'.



Research

Gu, X., Kannan, P. K., & Ma, L. [Selling the premium in freemium](#). *Journal of Marketing* (October 2018).

Universities: University of Maryland

[Full Ariyh research summary](#)

PRO TIP: Strongly emphasize the benefits of your paid plans

People are strongly anchored to "free". In one study, even [a minimal increase from €0.00 to €0.49 caused a drop of 8% of users](#). To counter this, be very clear about the added benefits of paid plans and make sure that people strongly value them.

“ 3.

LTV and MRR optimization

Preventing churn

Give gifts to long-time customers to increase their LTV



Recommendation

- Give free gifts to customers who have been with you long enough. For example, 20% higher extra bandwidth, for free. Make clear the unexpected reward is a thank-you gift for being a loyal customer, with no strings attached.
- Rewards reduce the churn of long-time customers but also slow down some of their natural increases in spending. Put together, it's still more profitable to reduce their churn.
- Do not give these to recent customers. This would encourage them to churn and look for more free rewards elsewhere.



How it works

- About 70% of customers are “inert non-switchers” (i.e. they are happy with the status quo and can't be bothered to look for another solution). Free gifts reawaken them to reconsider their options.
- For customers who have been with you for a while, the free gift is a way to justify and reinforce that they made the right decision in choosing you. This outweighs the likelihood that they will consider other options, and reduces their churn.
- Customers who have been with you for a long enough time have invested significant resources in you and don't want to regret doing so. They want to justify their previous decisions and dread the idea they could have made a mistake. This outweighs the likelihood that they will consider other options. Rewards reinforce their conviction that they made the right choice choosing you, reducing their churn.



Research

Henderson, C. M., Steinhoff, L., Harmeling, C. M., & Palmatier, R. W. [Customer inertia marketing](#). *Journal of the Academy of Marketing Science* (September 2020).

Universities: University of Oregon, University of Rostock, Florida State University, and University of Washington.

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Create an auto-enrolling simple loyalty program



Recommendation

- Create an automated loyalty program into which new customers are automatically enrolled when they subscribe (e.g. Get 2 months of free when you reach 12 months of subscription). These goals and rewards are clear to the customer in advance.
- In experiments, this increased projected customer lifetime value by 29.5% over 5-years. 80% of the increase is due to fewer customers leaving, not increased sales.



How it works

- People who are part of a loyalty program feel like the company is investing in their customers. This boosts a sense of connection.
- People also experience a quest for completeness. For example, if someone needs to stay with the company for a year to unlock a badge, the idea of reaching a milestone could be even more satisfying than the reward itself.



Research

Gopalakrishnan, A., Jiang, Z., Nevskaya, Y., & Thomadsen, R. Can Non-tiered Customer Loyalty Programs Be Profitable?. *Marketing Science* (March 2021).

Universities: University of Oregon, University of Rostock, Florida State University, and University of Washington.

[Full Ariyh research summary](#)

Offer generous downgrades to customers going through a hard time



Recommendation

- Many customers will churn because of a budget contraction (e.g. lower marketing budget this year and need to cut costs). However, this contraction is often temporary (e.g. in 3 years the budget will be higher than it was before).
- Do everything you can to detect this type of churn (e.g. questionnaire asking the reason for canceling their subscription) and instead of letting them churn offer an aggressively discounted temporary downgrade. Make sure it's clear that it's temporary (e.g. "To help you adjust to your budget requirements of this year"), and monitor for abuse.
- When their budget increases again, you will be one of the few subscriptions left in their arsenal, and they will be more likely to increase spending with you, rather than buy from others.



How it works

- When people have to re-evaluate their preferences, they reassess what gives them the most value.
- If they cut something out, they will likely not return to it. If they stick to it, they will be even happier with it (and self-justify their choice).
- These updated preferences remain the same even when their budget grows again.



Research

Ross, G. R., Meloy, M. G., & Carlson, K. A. [Preference Refinement After a Budget Contraction](#). *Journal of Consumer Research* (June 2020).

Universities: Texas Christian University, The Pennsylvania State University, and College of William and Mary.

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PRO TIP: When regaining churned customers, start with a low price first

In a similar way, be [generous with customers who return to you](#) after churning. Give them special low prices at the beginning. Once they are established customers again, you will be able to increase prices to higher than average and recoup your investment.

Learn how to manage churn and retention

- The main [drivers of churn](#) are typically competition, new investments, and negative experiences. Switching costs (e.g. the effort to move to a new platform) are typically lower than you would expect. To prevent churn, monitor what your competition offers, prevent negative experiences, and invest in the relationships with the customer.
- Customer satisfaction (e.g. ease of use, privacy protection) and perceived value (e.g. clear messaging and actual value of the product) are the [major drivers](#) of customer loyalty and reduced churn.
- To reduce churn, strengthening your branding and messaging (e.g. reminding about your key benefits) is [typically more effective](#) than reducing your prices compared to your competitors.
- Too frequent promotions and discounts [can backfire](#) and increase the likelihood of customers churning and switching to a competitor.
- If you have sophisticated enough analytics, target retention offers to specific customers that are both [profitable enough and likely to churn](#). You can [use this code developed by researchers](#) to optimally target offers.
- Remember it's easy to focus too heavily on churn prevention rather than improving new customer acquisition. Higher profitability is often [more tied to higher acquisition](#), than higher retention.



Affiliates

Optimize your affiliate payment model



Recommendation

- If your affiliate program is highly targeted and curated, where you onboard each affiliate individually and have a close relationship with each, use a pay-per-lead model (a fixed fee for each lead, which you will then nurture and try to convert into a customer). It's more predictable and profitable.
- If your affiliate model is one-to-many, where the barriers for joining are low (e.g. just sign up and start sharing the affiliate link), use a pay-per-conversion model (affiliates are paid only for who actually converts to a customer). This prevents affiliates from sending fake leads and maximizes affiliate revenue.



How it works

- Pay-per-lead is preferable in one-to-one agreements because it removes the distortion in pricing (i.e. the referrer gets the same payout no matter the spend of the new lead), leading to more efficient outcomes.
- However, in one-to-many arrangements, pay-per-conversion helps prevent fake leads and aligns both parties' interests by eliminating price distortion.



Research

Libai, B., Biyalogorsky, E., & Gerstner, E. [Setting Referral Fees in Affiliate Marketing](#). *Journal of Service Research* (May 2003).

Universities: Israel Institute of Technology, University of California

PRO TIP: Don't let affiliates clash with your own marketing efforts

Set and enforce clear rules when it comes to how affiliates can promote you. For example, affiliates shouldn't be allowed to advertise you directly in search ads (e.g. Google, Bing) if you are doing so yourself - they will [cannibalize your efforts](#). The same goes for your SEO efforts. Instead, give them tracking links that they should use in more specific channels, usually ones that they own (e.g. their own blog or newsletter).

Protect your brand from deceptive affiliates



Recommendation

- Require partners and influencers to be transparent about the affiliate partnership.
- Monitor and identify patterns of fraudulent activities of your affiliates (e.g. overnight spikes in follower counts and disproportionate followers-to-engagement ratios), and eliminate those affiliates from your program.
- At first glance, it may seem advantageous to keep working with affiliates that are not fully compliant (“it’s traffic, after all”) but they tend to hurt your brand and rarely lift sales.



How it works

- Some influencers inflate engagement metrics using fake accounts or by participating in engagement pods that exchange fake interactions (e.g. likes, follows) to boost their perceived influence. Higher engagement metrics can lead to better compensation in partnership deals, motivating influencers to manipulate these metrics.
- Only less than 10% of influencers disclose affiliate links. As people become more aware of the commercial nature of influencer posts, undisclosed affiliations decrease trust and engagement, negatively impacting brand perception and reputation.



Research

Mangiò, F., & Di Domenico, G. [All that glitters is not real affiliation: How to handle affiliate marketing programs in the era of falsity](#). *Business Horizons* (2022).

Universities: University of Bergamo, Cardiff University

PRO TIP: Set up workflows to minimize affiliate marketing risks

Avoid risks and manage [your affiliates effectively](#) by creating clear guidelines for creators (e.g. providing messaging run downs, a set of rules, keywords) and opting for in-house management over outsourcing to agencies.

PRO TIP: Choose partners and influencers with a strong audience fit

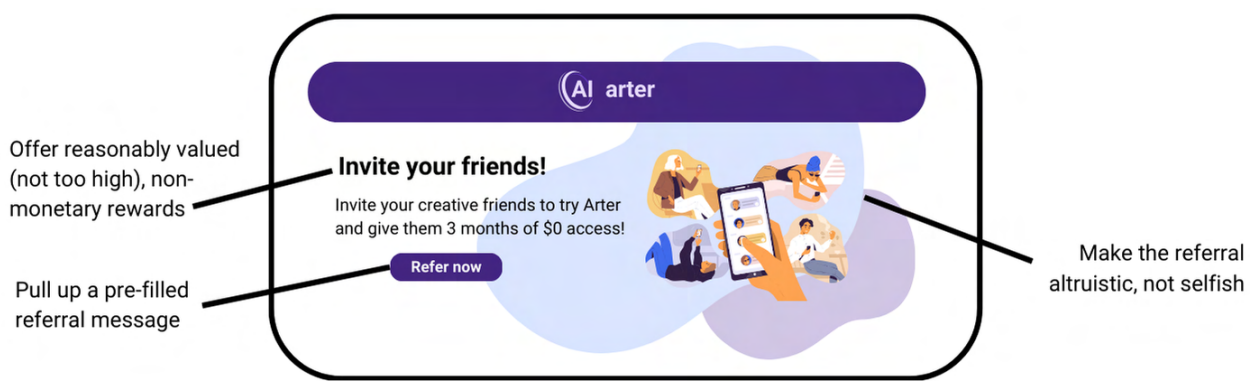
Language on social media posts can impact the success of affiliate marketing campaigns. Work with creators that use [engaging, authentic, and confident language](#). This will boost engagement and click-through rates.



Referrals

This is an example mockup of an optimized SaaS Referrals page.

Make sure to apply these insights when designing your referral program to optimize conversions and effectiveness.



Make your referrals altruistic



Recommendation

- When asking customers to refer you, use an altruistic framing of “Invite a friend to give them a month for free” rather than a selfish “Invite a friend to get a month of free access”. In experiments, the likelihood of referrals increased by 60% and was up to 86% more effective.
- Target your referral campaign to loyal and/or highly satisfied customers (e.g. those who made repeat purchases), and try to ask for a referral right after the customer has purchased your product.



How it works

- Word-of-mouth is by nature something people do altruistically because they think the other person would enjoy it. When there’s a selfish financial incentive, the two incentives contrast and damage each other.
- Instead, describing a referral as something altruistic reduces the senders’ guilt of getting a financial reward.
- It also encourages better targeting by the senders, which leads to higher conversion rates.



Research

Jung, J., Bapna, R., Golden, J. M., & Sun, T. [Words Matter! Toward a Prosocial Call-to-Action for Online Referral: Evidence from Two Field Experiments](#). *Information Systems Research* (May 2020).

Universities: Temple University; University of Minnesota; and University of Southern California.

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PRO TIP: Employees are a strong source of altruistic referrals

Employees are already likely to refer their peers, and having a program in place will [boost the chances of it happening even more](#). Design the referral so that the receiver will receive the rewards (e.g. “As an employee you can gift 3 months off to a friend that joins”), it can boost the referral rates up to 22%.

Pre-fill your referral messages to make them more effective



Recommendation

- Create workflows to automate and prefill your referral messages. For example, a CTA “invite friend” button pulls up a prefilled email window with the message “I’ve been using the Grammaroo editor for over 6 months and I love it! I think you’ll find it useful too”.
- You can prefill the message in one of two ways, with different effects (don’t do both or it will backfire):
 - Say that the sender has already bought the product, this increases conversions of the receiver by 11%.
 - Explain that there is an incentive for sharing this referral (e.g. “Buy and then share a referral to get a reward”) to increase further referrals by 53%.



How it works

- Social learning that a friend has already made a purchase makes people more comfortable in buying a product themselves.
- Further referrals are motivated by a potential gain and incentive they could then experience themselves.



Research

Sun, T., Viswanathan, S., & Zheleva, E. [Creating social contagion through firm-mediated message design: Evidence from a randomized field experiment](#). *Management Science* (August 2020).

Universities: University of Southern California, University of Maryland, and University of Illinois.

[Full Ariyh research summary](#)

Avoid money as a referral reward



Recommendation

- Avoid using monetary rewards to customers who refer new buyers (e.g. "Invite a friend and get \$20").
- Instead, offer non-money rewards such as a free extra product (e.g. a free ebook, more data storage).
- It doesn't have to be one of your own products (e.g. you can offer partner gifts).



How it works

- People recommend products to friends because they want to help them.
- Money rewards change an altruistic action to a 'business transaction'. It creates distance and suspicion.
- Non-money rewards also change social dynamics (the recommender is gaining something instead of just being nice), but the impact is not as drastic as when money is involved.



Research

Jin, L., & Huang, Y. [When giving money does not work: The differential effects of monetary versus in-kind rewards in referral reward programs.](#) *International Journal of Research in Marketing* (October 2013).

Universities: Fudan University

[Full Ariyh research summary](#)

Keep referral rewards reasonably small



Recommendation

- Choose the size of your referral incentives carefully. You don't want them to be too high relative to the value of your product (e.g. a \$50 value reward is likely too much for a SaaS that costs \$5/month; a referral with a value of \$5-\$10 would be more reasonable) or in absolute value (even if your SaaS is \$500/month, a \$1,000 reward is likely too high).
- Large referral rewards are unprofitable in most cases. They cause a huge boost to your referral rate (+750% in one experiment) but lower the average lifetime value of newly referred customers to a level that in most cases is unprofitable (-48% in the experiment).



How it works

- Referred customers are more valuable than average because people tend to refer people who are an excellent fit for the product.
- Large rewards motivate to refer beyond that group, to people who are likely not really interested in your product - attracting bad-fit, less committed, price-conscious customers who are less profitable.



Research

Wolters, H. M., Schulze, C., & Gedenk, K. [Referral reward size and new customer profitability](#). *Marketing Science* (October 2020).

Universities: University of Hamburg, Frankfurt School of Finance & Management, and Xing

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Badly targeted referrals can increase your churn



Recommendation

- Encourage people to only refer friends who are likely to buy your product. For example, remind who it's for (e.g. "Recommend Ariyh to other research-based marketers" or "Invite other bookworms to Goodreads").
- If they recommend it to someone who won't follow the recommendation (e.g. because the product is not a good fit for them), the referrer will be up to 26% more likely to churn in the future.



How it works

- People offer recommendations because they think their opinions are sound.
- When the recommendation is rejected, people's confidence in knowing the best products suffers and they may switch to alternative options to "regain expertise".



Research

Hall, M. J., Hyodo, J. D., & Kristofferson, K. [How rejected recommendations shape recommenders' future product intentions](#). *Journal of Consumer Psychology* (December 2023).

Universities: Oregon State University, Western University & Ivey Business School. Western University

[Full Ariyh research summary](#)



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Thomas McKinlay,
Founder of Ariyh